

How We Tested

The system we tested the drives on was a Pentium 4, 2.8 GHz on an Intel D850MV motherboard with integrated sound card and 512 MB RD RAM, a GeForce3 display card, a 40 GB Seagate Barracuda as primary master and a Samsung 52X CD-ROM as primary slave, running Windows XP Professional. All drives were then connected as secondary masters, and their performance in the various test categories was noted. Performance accounted for 90 per cent of the drives score, while the remaining 10 per cent was based on less important details such as packaging, half-height, etc. The value for money was calculated as a scaled ratio of the scores, obtained in the features and performance tests, divided by the price of the drive. To make things easier, we divided the drives into two categories—CD-Writers and combination drives (CD read, write, rewrite and DVD read).

CD read, write and re-write tests

These tests are common for both categories. We used Nero CD Speed and SiSoft Sandra for the read speed test.

Nero CD Speed: This software is used to test the data transfer rate, access or seek time, CPU usage time, the burst rate of the drive, drive motor spin up and spin down, and finally, the DVD eject and load time. Tests that are related to time are measured in milliseconds; data transfer is either measured in KBps or MBps while drive speeds are measured in terms of X. We used the same data CD on all the drives. We also did a Digital Audio Extraction

(DAE) quality test of the drives. In this test, the software extracts audio sectors from three different locations on the CD and stores it on the hard drive. The CD is then read again, and compared to the extracted audio on the hard drive. A pressed audio CD was used to get the DAE quality results. Nero CD Speed assigns scores ranging from 0 to 10 for the DAE test, with 10 given for no difference in the files.

SiSoft Sandra: Here, we measure the Sequential and Random Read speeds of the drive, as well as the average access time. The media used here is the same data CD that was used for the Nero test. Higher results in the Sequential and Random read speed tests, and lower access times, attracted more points.

Audio Extraction Speed Test: We used a pressed audio CD and extracted the first, middle and last tracks from it audio CD. The criteria considered here were the time taken by the drive to rip the tracks and convert them to .wav format. The audio CD used was the same for all the drives. The quality of the track is not given much importance; however, the presence of audible hisses, clicks and pops was noted, and subtracted from its score.

CD writing tests: This is a complete battery of tests which was common for both categories of drives. The tests comprised writing data sequentially (single file test), random writing (assorted write test), burning an image file, a CD-to-CD copy test and writing an audio CD. The time taken for each is measured and the points are awarded accordingly. The Samsung 52X CD-ROM on our test bed was used in the

CD-to-CD copy test for all drives. We also used the fastest media available—48X.

CD re-writing tests: We did two tests here—the assorted write speed test and the CD-RW full erase test, and the time taken for each was noted. We had to make do with 24X re-writable media, as this is the fastest available.

Copy Protection Tests: This is a new category of tests, which comprises of making a back up of a game or data CD that is copy protected. This is important as it is a very gruelling task for the drives, and helps us test the lens quality. Points were awarded to the drives which were able to perform this test without any hitches.

DVD read tests

For the DVD performance test, we used three different software; each covering a particular aspect of the drive's read performance.

Nero DVD Speed: This test was the same as the CD tests, except we used a pressed DVD. The same DVD was used to test all the drives. Also, the maximum number of DVD formats supported was also considered—and scores were awarded accordingly. Points were given for access time, burst rate, spin-down time and disc recognition times.

SiSoft Sandra: This test was performed twice for the combo drives, to measure both CD and DVD read speeds.

DVD Decrypter: This software is used to extract a .vob file from the DVD to the hard drive, and the time taken is measured. This test was only applicable for combination drives. Lower extraction times were given higher scores.

Plextor Premium

Does the thought of fitting 1 GB of data on a 700 MB CD interest you? Well, this Plextor drive lets you do just that. Not only can you fit 1 GB of data on a normal 80 minute, 700 MB CD-R, you can password-protect your data as well. If you try playing this password-protected CD on any other computer, you will get a disc-read error. This can be controlled by downloading the SecuView software and running it on the system where you want to view the CD.

Plextor calls this feature GigaRec, and the password-protection feature is called SecuRec. The maximum capacity of a disc is dependent on the type of disc and the GigaRec rate. All this looks rosy, but there are a few catches—the discs burned, using the GigaRec feature are not read correctly in some drives.

whooping 509 seconds. The MSI finished the fastest, with 448 seconds. Moving to the assorted data CD

re-writing test, the BenQ drive took the top spot with a speed of 217 seconds. The MSI drive was the first to erase the re-writable disc, finishing in just 218 seconds. The AOpen finished last, taking 312 seconds for both the writing and erasing.



The AOpen CRW 5224 is a slow re-writer

In the EFM test, the drives were supposed to make an exact copy of an EFM encoded, copy-protected disc. The BenQ failed this

test, while all the others passed with flying colours.

There were a few annoying things that we came across while testing. The AOpen, Sony and MSI drives are set to run at a fixed speed of 40X, with 48X for Sony. You need to press and hold the eject button to make the drive write at faster speeds. Also, the BenQ drive was temperamental and